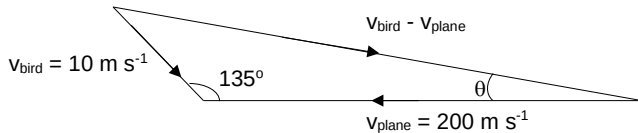


VJC Prelim 2020 H2 Physics Paper 3 – Suggested Solutions

1(a)



(b) Speed of bird relative to plane = $\sqrt{10^2 + 200^2 - 2(10)(200)\cos 135^\circ}$
 $= 207 \text{ m s}^{-1}$

$$\frac{\sin \theta}{10} = \frac{\sin 135^\circ}{207}$$
$$\theta = 1.96^\circ$$